Faculty of Computing & Information Technology

CC-112L : Programming Fundamentals **Lab Instructor**: Dr. Muhammad Farooq

BSAI/IT Morning/Self 2024
LAB – 01

Date : 1/Friday/2026

The objective of this Lab is to:

- Pseudocode
- Integer and Double data types
- Operators, Arithmetic

Question 1. Print your name, rollNumber, Subject, University name, and Teacher Name. (Your output should be in a presentable way).

Question 2: Write pseudocode for a system that:
Takes:
Number of items
Price per item
Calculates:
Total bill
5% service charges
Final payable amount
Display all results clearly

Question 3: Write a pseudocode algorithm that:
Takes a 4-digit number
Displays digits in the following order:
Order is not simple reverse
Use only / and %
SAMPLE:
Input : 7531
Output : 1 - 7 - 3 - 5

Question 4: Write pseudocode that:
Takes a 4-digit integer as input
Calculates:
Sum of all digits
Difference between first and last digit
Displays both results

Question 5: Design pseudocode (sequence only) that:
Takes input for 5 students: name and marks in 3 subjects (Math, Science, English).
Calculates for each student:

Total marks
Average marks
Then calculates class averages for Math, Science, and English.
Finally displays:
Each student's name, total, average
Class averages for each subject

Question 6: A student's final grade is calculated as:
Assignment 1 (weight 20%), Assignment 2 (weight 30%), Exam (weight 50%).
Write pseudocode (sequence only) that:
Takes scores for A1, A2, and Exam (out of 100).
Calculates weighted score =
(A1×0.2 + A2×0.3 + Exam×0.5).
Displays each component score and final weighted score

Question 7: The gravitational potential energy of an object is calculated using the formula: PE = m x g x h
Where :
m = mass of the object (in kilograms)
g = gravitational acceleration (9.8 m/s^2)
h = height above the ground (in meters)

Question 8: Design an algorithm that:
Takes input:
Number of burgers ordered
Number of fries ordered
Number of pizzas ordered
Prices:
Burger = $5 each
Fries = $3 each
Pizza = $8 each
Processing:
Calculate total before tax.
Add 8% tax to the total.
Output : Display itemized cost and final cost.

Question 9 : A delivery service charges a base fee of $10.
Input: Take the weight of a package in kilograms and the distance in kilometers.
Process: The cost is the base fee + ($2 per kg) + ($0.50 per km).
Output: Display the total delivery cost.

Question 10: Write pseudocode that takes an amount in USD from a user.

Process: First, deduct a fixed $5 handling fee. Then, convert the remaining amount to Euros (assume 1 USD = 0.92 Euros).

Output: Display the final amount in Euros

Question 11. Write a pseudocode that stores the integers 50 and 100 in variables, and stores the sum of these two in a variable named total.

Question 12. Write pseudocode to:

- Input distance (in kilometers)
- Input time (in hours)
- Calculate speed
- Display the speed (Formula: Speed = Distance / Time)

Question 13. Write pseudocode to:

- Input marks of two subjects
- Calculate average marks
- Display average marks

Question 14. **Electricity Bill Calculation**

Write pseudocode to:

- Input number of units consumed
- Input rate per unit
- Calculate total bill
- Add fixed meter charges
- Display final bill

Question15: Given the following pseudocode:

```
integer a, b, c
a = 20
b = a / 6
c = a % 6
a = b + c
b = a * 2
c = b / 5
a = a + c
b = a % 4
output a
output b
output c
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a) Find output b) Explain each step briefly

Question 16. Write pseudocode to output the following pattern:

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Question 17. Write pseudocode to solve the following problem:

A stopwatch counts time only in total seconds. You need to convert a large input of seconds into a standard digital clock format (Hours, Minutes, and Seconds).

* Input: One integer representing total seconds (e.g., 3672).

* Process: Use integer arithmetic to calculate the exact hours, remaining minutes, and remaining seconds.

* Output: Print three separate integers: Hours, Minutes, Seconds.

* Example:

Input 3672

Output: 1 Hour, 1 Minute, 12 Seconds.

Question 18. Write pseudocode to help the ATM work rightly:

An ATM needs to dispense a specific amount of money using the fewest possible notes. The available notes are 500, 100, and 10. (Assume the user requests a multiple of 10).

* Input: One integer amount (e.g., 1630).

* Process: Calculate how many 500s, then how many 100s, then how many 10s are needed. (Note: Since you cannot use loops, you must calculate the remainder passed down from one denomination to the next).

* Output: Three integers representing the count of 500s, 100s, and 10s.

* Example: Input 1630 Output: 3 (500s), 1 (100s), 3 (10s).

Question 19. Write pseudocode to find out how much paint is required:

A particular brand of paint covers **340 square feet per gallon**. Write a program to determine and report approximately how many gallons of paint will be needed to paint **two coats** on a wooden fence that is **6 feet high** and **100 feet long**.

Question 20. Write pseudocode to solve the following problem:

Suppose an employee gets paid every two weeks and earns $2,200 each pay period. In a year, the employee gets paid 26 times. Write a program that defines the following variables:

payAmount This variable will hold the amount of pay the employee earns each pay period. Initialize the variable with 2200.0.

payPeriods This variable will hold the number of pay periods in a year. Initialize the variable with 26.

annualPay This variable will hold the employee's total Annualpay, which will be calculated.

The program should calculate the employee's total annual pay by multiplying the employee's pay amount by the number of pay periods in a year and store the result in the annual Pay variable. Output the total annual pay.